

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
O-LEVEL · COMBINED SCIENCE

COMBINED SCIENCE 5006/2

PAPER 2 November 2022

2 hours

PAPER 2 — STRUCTURED QUESTIONS — ANSWER ALL — 2 HOURS

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers. This script is unique to November 2022 5006/2.

Additional materials: Answer all questions. The number of marks is given in brackets [].

INSTRUCTIONS: Write your answers in the spaces provided. You may use a calculator.

- 1.** (a) Name two structures found in a plant cell but not in an animal cell. [2]<<
>>(b) State the function of the cell membrane. [2]<<
>>(c) Describe how to prepare a slide of onion epidermis to view under a light microscope. [3]<<
>>(d) Explain why muscle cells contain many mitochondria. [3] **[10]**
- 2.** (a) Write the balanced symbol equation for photosynthesis. [3]<<
>>(b) Name the green pigment and the organelle where it is found. [2]<<
>>(c) State two factors that can limit the rate of photosynthesis. [2]<<
>>(d) Describe a test for starch in a destarched leaf that has been exposed to light. [3] **[10]**
- 3.** (a) What is a pathogen? [2]<<
>>(b) Distinguish between an antigen and an antibody. [2]<<
>>(c) Explain how a vaccine can protect a person against a disease. [4]<<
>>(d) Suggest why antibiotics are not used to treat a viral infection such as influenza. [2] **[10]**
- 4.** (a) State where sperm are produced and where fertilisation occurs in humans. [2]<<
>>(b) Name two male secondary sexual characteristics. [2]<<
>>(c) Describe one barrier method of contraception and how it works. [3]<<
>>(d) Explain how the placenta is important to the fetus. [3] **[10]**

5. (a) What does the group number of a main-group element tell you? [2]<<
>>(b) What does the period number tell you? [2]<<
>>(c) Explain why noble gases are unreactive. [3]<<
>>(d) State how the reactivity of Group 1 metals changes down the group, and why. [3] **[10]**
6. (a) Write a word equation for hydrochloric acid reacting with calcium carbonate. [3]<<
>>(b) Describe the test for carbon dioxide. [2]<<
>>(c) Universal indicator is added to the acid before the carbonate. Describe the colour change as the acid is used up. [2]<<
>>(d) Explain why farmers add lime (calcium carbonate) to some soils. [3] **[10]**
7. (a) Define average speed. [2]<<
>>(b) A learner runs 60 m in 5 s. Calculate the average speed. [3]<<
>>(c) A car travels at constant velocity. What is the resultant force on the car? Explain. [3]<<
>>(d) Name the instrument used to measure time in this experiment. [2] **[10]**
8. (a) State the rule for attraction and repulsion of magnetic poles. [2]<<
>>(b) Name two magnetic materials. [2]<<
>>(c) Describe how to plot the magnetic field of a bar magnet using a plotting compass. [3]<<
>>(d) Explain how an electromagnet can be made stronger. [3] **[10]**

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).